

Automated Coin Detection and Extraction

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Made with GAMMA

The Challenge

Textured Backgrounds

Wood grain complexity

Uneven Illumination

Shadows & lighting

Metallic Reflections

Specular highlights



Processing Pipeline



Grayscale

Blur

Threshold

Morphology

Extraction

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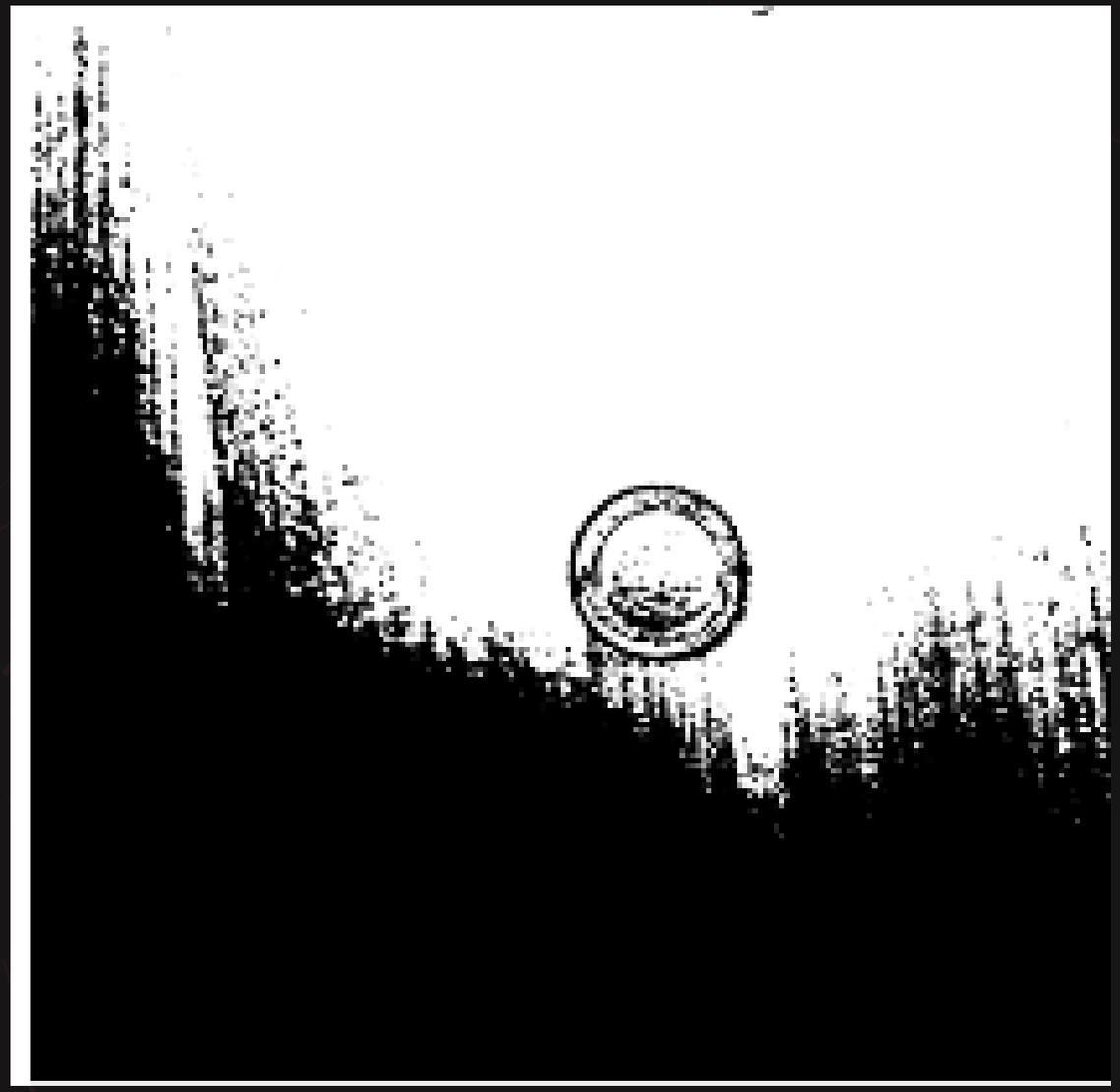
Adaptive Thresholding

Local vs. Global Analysis

Dynamic threshold per region

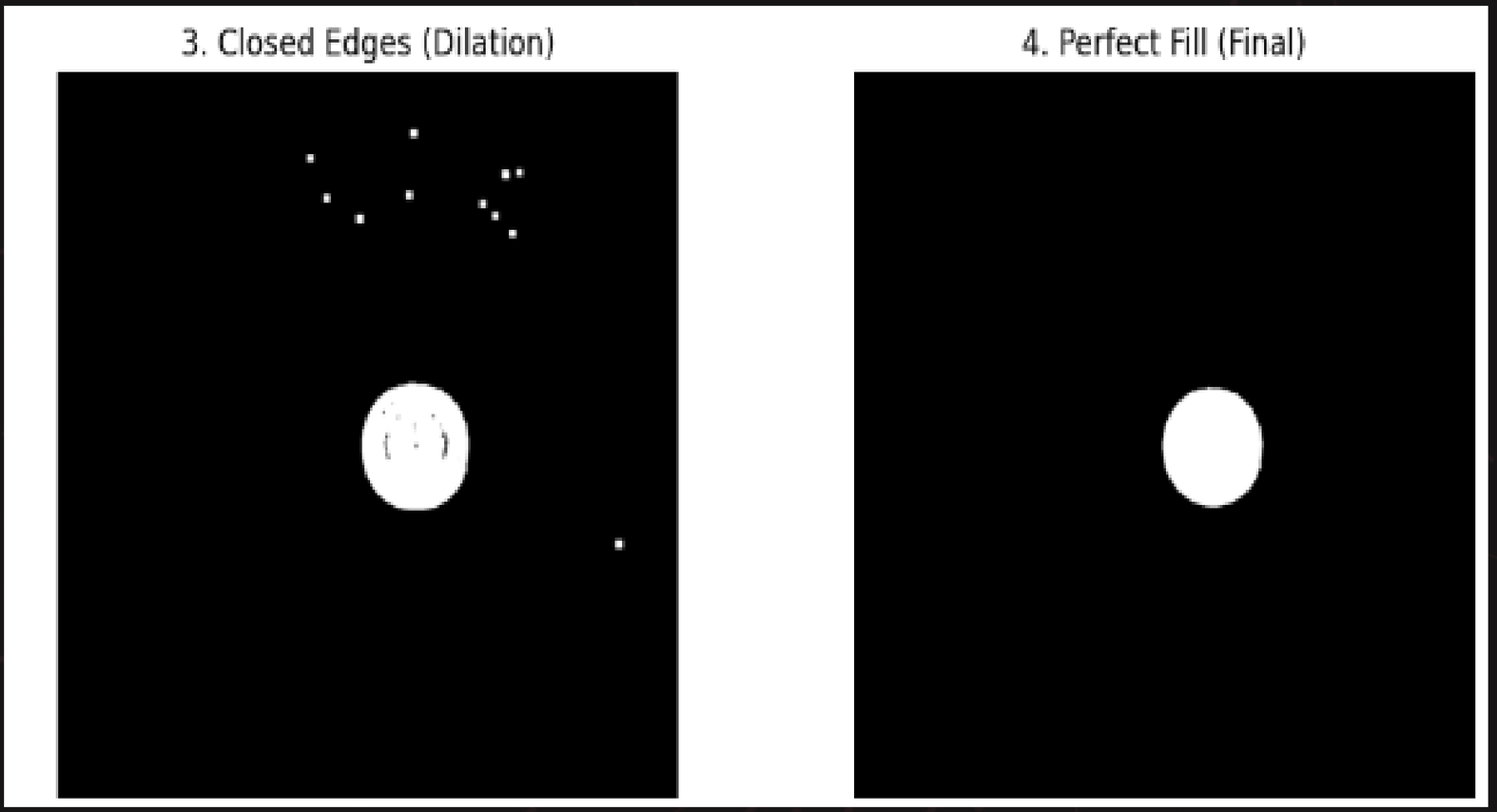
Overcoming Shadows

Local brightness compensation



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Morphological Closing



Dilation + Erosion

Fill gaps, smooth edges

Fixing Broken Edges

Reconnect fragmented boundaries

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Implementation

1

Python 3.13

Modern interpreter

2

OpenCV & NumPy

Core libraries

3

Gaussian 15×15

Blur kernel size

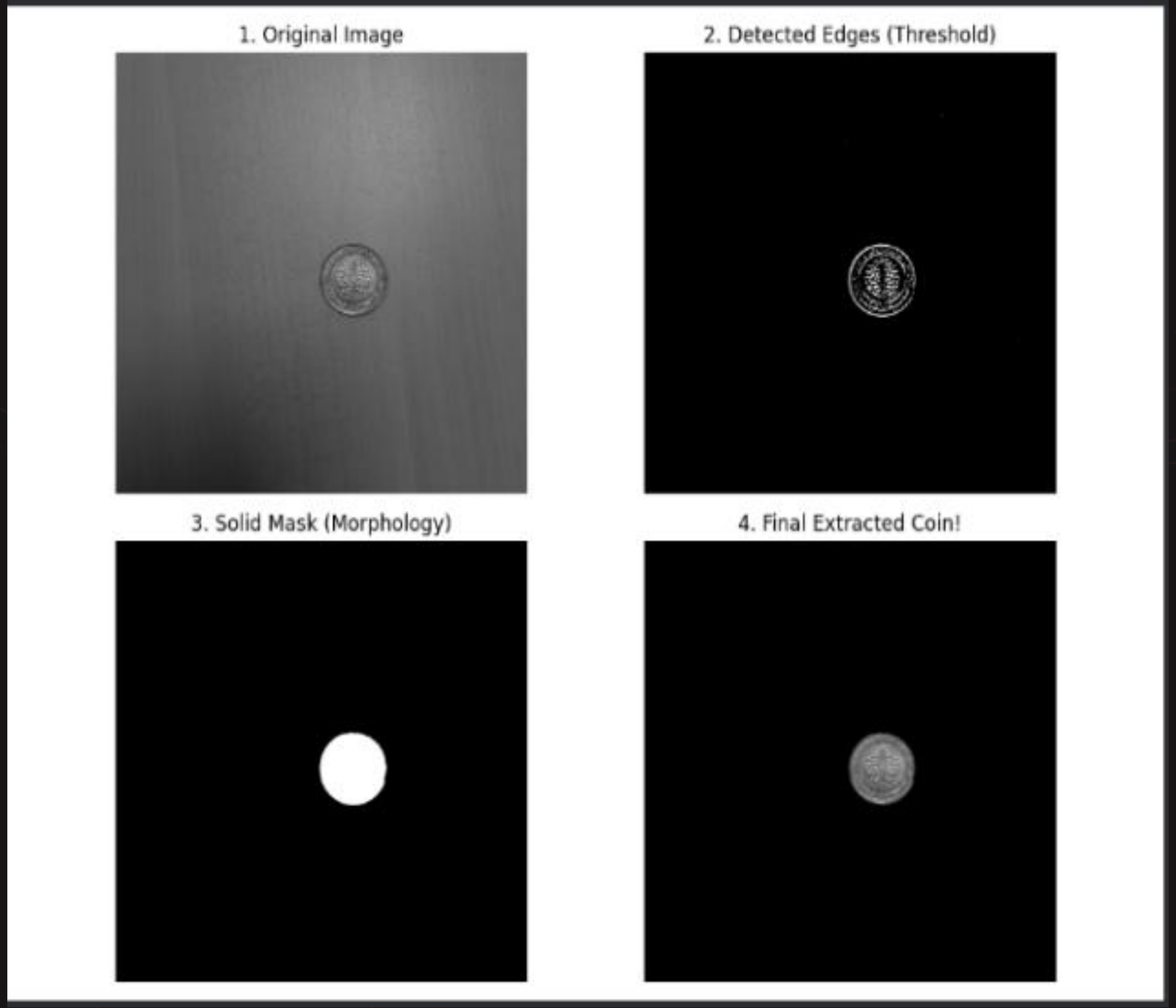
4

Ellipse 25×25

Morphology kernel

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Results



21.10
MSE Score

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Conclusion & Future Work

Successful Extraction

Coins isolated

Future: Automated Tuning

Dynamic parameters

Hough Circle Transform

Shape detection

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